

Hafiz Muhammad Usama Saleem

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Portfolio: [Website](#) | LinkedIn: [Profile](#) | GitHub: [Profile](#)

Machine Learning Engineer specializing in LLMs, RAG systems, and voice automation. Experienced in building AI agents, fine-tuning models, and deploying scalable GenAI applications using LangChain, CrewAI, and FastAPI. Passionate about creating intelligent, cost-effective AI systems that drive measurable business impact.

EDUCATION

MSc in Data Science

FAST NUCES, 2025

Postgraduate Diploma in Data Science and Artificial Intelligence

NED University, 2024

Bachelor of Engineering in Electrical

NED University, 2020

WORK EXPERIENCE

Machine Learning Engineer at PixelEdge Solutions (Pvt.) Limited (October 2023 – Current)

Voice AI & LLM Automation

- Built an AI Voice Assistant using Twilio, Speech-to-Text (STT), TTS, and LLMs (OpenAI + open-source) to automate customer interactions.
- Delivered human-like conversations, reducing dependence on support agents and improving response efficiency.
- Deployed on AWS with FastAPI, ensuring real-time performance, scalability, and high availability.

GenAI – RAG-Based Intelligent Systems

- Developed a RAG pipeline using LangChain, OpenAI embeddings, and OpenSearch to answer queries from sustainability and policy documents.
- Integrated with FastAPI + ChatGPT API, enabling context-aware, natural responses.
- Designed modular REST APIs to expose retrieval + generation capabilities across internal platforms.

Classical ML – Fintech Data Automation & KPI Prediction

- Built ML workflows for financial data processing, feature engineering, and KPI forecasting, reducing manual financial reporting.
- Generated automated Excel-based insights from raw financial inputs, improving business decision-making.
- Used Python, Scikit-learn, Pandas, NumPy, Flask/FastAPI for full pipeline development & deployment.

Agentic AI Systems – Autonomous Research Workflows

- Built multi-agent AI systems using SmolAgent, CrewAI, and LangGraph, powered by Gemini LLM.
- Designed agents for autonomous web search, information extraction, reasoning, and report generation.
- Implemented task orchestration for scalable enterprise research automation.

Trainee Data Scientist at Techma Zone (January 2023-October 2023)

- Performed data preprocessing, cleaning, and feature engineering on real-world datasets to prepare for analysis and modeling.
- Conducted exploratory data analysis (EDA) and visualized insights using tools like Matplotlib, Seaborn, and Plotly.
- Collaborated in cross-functional projects to apply machine learning algorithms and evaluate model performance using industry-standard metrics.

TECHNICAL SKILLS

- **GenAI:** LangChain, LLM Integration, Prompt Engineering, RAG, Vector Databases
- **Agentic AI Frameworks:** CrewAI, SmolAgent, LangGraph
- **Machine Learning:** Supervised & Unsupervised Learning, Transformers, Model Evaluation
- **Deep Learning:** PyTorch, TensorFlow, Keras
- **Natural Language Processing:** NLTK, spaCy, Text Classification, Summarization
- **Voice AI:** Whisper, Twilio, Speech-to-Text (STT), Text-to-Speech (TTS)
- **Programming Languages:** Python, SQL, JavaScript (basic understanding)
- **Databases:** PostgreSQL, MySQL
- **Cloud & DevOps:** AWS, Docker, GitHub Actions, CI/CD
- **Model Deployment & APIs:** Flask, FastAPI, REST APIs, Authentication & Authorization
- **Frameworks & Tools:** Scikit-learn, Hugging Face, Jupyter Notebook, Git, Pandas, NumPy
- **Data Engineering & Visualization:** ETL Pipelines, Semantic Chunking, Matplotlib, Seaborn, Plotly

PERSONAL PROJECTS

Mini AI Concierge (Guest Support Bot) [GitHub](#) | [Loom Demo](#)

Built an LLM-powered chatbot using RAG architecture to automate guest support for vacation rentals. Integrated document retrieval and fallback escalation logic using FastAPI, FAISS, and Gemini.

Agentic Travel Companion [GitHub](#) | [Loom Demo](#)

AI-powered Streamlit application that transforms travel planning into an interactive, personalized experience. Utilizes multiple specialized AI agents to assist with destination selection, budgeting, and trip itineraries using CrewAI.

Stock Price Prediction Model [GitHub](#)

Developed a logistic regression model to predict stock price movement based on financial indicators, using exploratory data analysis and feature selection techniques.

CERTIFICATION

Generative AI with LLMs | Coursera

Complete Machine Learning & Data Science Bootcamp | Udemy

100 Days of Code: The Complete Python Pro Bootcamp | Udemy

Mathematics for Machine Learning and Data Science Specialization | Coursera

CORE COMPETENCIES

Problem-Solving | Communication | Adaptability | Time Management

LANGUAGE SKILLS

English | Urdu